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Towards Improving Privacy of Synthetic DataSets

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Abstract. Recent growth in domain specific applications of machine learning can be attributed to availability of realistic public datasets. Real world datasets may always contain sensitive information about the users, which makes it hard to share freely with other stake holders, and researchers due to regulatory and compliance requirements. Synthesising datasets from real data by leveraging generative techniques is gaining popularity. However, the privacy analysis of these dataset is still a open research. In this work, we fill this gap by investigating the privacy issues of the generated data sets from attacker and auditor point of view. We propose instance level Privacy Score (PS) for each synthetic sample by measuring the memorisation coefficient α_m per sample. Leveraging, PS we empirically show that accuracy of membership inference attacks on synthetic data drop significantly. PS is a model agnostic, post training measure, which helps data sharer with guidance about the privacy properties of a given sample but also helps third party data auditors to run privacy checks without sharing model internals. We tested our method on two real world data sets and show that attack accuracy reduced by PS based filtering.

Keywords: Privacy Preserving Synthetic Data · Generative Adversarial Networks · Privacy Audit.

1 Introduction

Machine learning (ML) has been widely adopted in a variety of applications across multiple domains. Due to the maturity of large-scale training infrastructure [3], development tools and, the availability of large amounts of training data companies are building huge models with billions of parameters for commercial applications. The training data used to train these models often contain personal or sensitive data [1], which was aggregated from datasets that are used for purposes unknown to the data owner (i.e. the person itself). Unfortunately, end-users who play an active role as data donors lack visibility into how this data is being used/misused by the applications built on top of the data collected. A

growing number of AI incidents³ show an upward trend of data and model mis-uses, which include discriminatory hiring decisions, behavioral manipulations, and unethical systems that lack transparency. To gain the trust of users and avoid the privacy implications of data sharing for end users institutions [4–6] and companies [9–16] are adopting synthetic data [17] as a solution to avoid privacy issues.

Synthetic data as an idea was first adopted in designing various imputation methods in the early 90’s [17, 18]. Raghunathan et.al [32] formally defined a framework to define properties of synthetic data, which was further enhanced in multiple other researchers [19–22, 33, 34]. Many recent works leveraged traditional statistical techniques and machine learning models to generate synthetic data [23, 25, 26, 28–31].

Ideal synthetic data has properties such as handling large datasets with diverse data types (categorical, ordinal, continuous, skewed, etc.), preserving semantic relationships between variables [26] and, underlying algorithms are fairly generalizable and needs a little tuning. ML models trained on synthetic data may have similar performance as original data. Theoretical privacy guarantees can be achieved via ”ideal” synthetic data as a synthetic data record is not an actual data record of an individual but reflect the exact data distributions of original data.

In the context of data sharing, the majority of recent methods employ generative models [39, 40, 47, 50, 51] to obtain the synthetic data from real data. Popular generative modeling frameworks such as Generative Adversarial Networks (GANs [40]) and Variational Auto-encoders (VAEs) use real data during training to train a model capable of generating (or ”synthesising”) ”fake” data points. Quality metrics such as *fidelity* which measures how ”realistic” is the synthetic data when compared to real data and, *diversity* which measures how diverse the generated samples to cover the variability of real data are used to assess quality of the synthetic data.

Measuring the generalization property of synthetic data is very important and often ignored by synthetic data generators. Most of the proposed methods optimize on fidelity and diversity properties ignoring to measure or report, how well the synthetic samples capture the overall distribution including outliers of real data and, not just copies of the (real) samples of training data. Evaluating generalization in generative models is not easy [54, 55] when compared to discriminative models which are tested on held-out data.

From compliance and regulation view point, the General Data Protection Regulation (GDPR [42]) gives new rights to citizens regarding transparency around personal data. These include the right of access (knowing which data about the person has been stored) and the right to data erasure, also known as the right to be forgotten (having personal data deleted under certain circumstances). Privacy risk include the *risk of re-identification* and the *risk of inference* [35–37, 41]. When an attacker is able to identify and link individuals within a dataset or across multiple datasets, then these datasets pose the risk

³ <https://incidentdatabase.ai/>

of re-identification. These types of attacks can reveal the presence of user [43] or user data was used to train an ML model [44]. Deducing the private information of individuals with high confidence from the linked datasets or ML model fall under the risk of inference.

Motivated by the need from compliance, privacy of the end user and, growing popularity of synthetic data as sharing mechanism both in industry and academia, in this work we aim to investigate privacy properties of synthetic data generated by generative models. First, we propose a novel Membership Inference Attack (MIA), which uses the popular bagging technique to achieve attacker goal of identifying the membership of user in the model trained on synthetic data. Next, a instance level privacy score is calculated to filter the synthetic samples, which could compromise the privacy of the sensitive attributes in the synthetic data to protect from MIAs.

2 Background

We present a brief background on membership inference attacks and generative machine learning models.

Membership inference attack (MIA). is one of the most popular privacy attacks against ML models [73–79]. The attacker goal of membership inference (MI) is to determine whether a data sample x is part of the training dataset of a target model $\mathcal{T}$. We define a membership inference attack model $\mathcal{A}_{MemInf}$ as a binary classifier.

$$\mathcal{A}_{MemInf} : x, \mathcal{T} \mapsto \{member, non-member\} \quad (1)$$

MIA poses severe privacy risks, using a model trained on data samples of people with sensitive information an attacker can execute MIA to successfully infer the membership of person revealing the person’s sensitive information. To achieve MIA, Shokri et al [73] train a number of shadow classifiers on confidence scores produced by the target model with labels indicating whether samples came from the training or testing set. MIAs are shown to work in white-box [57] as well as black-box scenarios against a various target models including ML-as-a-service [73] and generative models [56]. Yeom et al. [59] explore overfitting as the root cause of MI vulnerability.

Similarly, a variant of MIA is attribute inference in which sensitive attribute can be inferred from the target model. This sensitive attribute is not related to the target ML model’s original classification task. For instance, a target model is designed to classify an individual’s age from their social network posts, while attribute inference aims to infer their educational background [71, 72]. Some studies [58, 59, 64] attribute membership inference attacks to the generalization gap, the over-fitting of the model, and data memorization capabilities of neural networks. Deep neural networks have been shown to memorize the training data [63, 69, 70], rather than learning the latent properties of the data, which means they often tend to over-fit to the training data. Weakness like this, in

some cases, can be exploited by an adversary to infer data samples' sensitive attributes.

Given a data sample x and its representation from a target model, denoted by $h = f(x)$, to conduct the attribute inference attack, the adversary trains an attack model $\mathcal{A}_{AttInf}$ formally defined as follows:

$$\mathcal{A}_{AttInf} : h \mapsto s \quad (2)$$

where s represents the sensitive attribute.

Generative Adversarial Network. (GAN) [40] is a type of adversarial training system, where two competing models are trained against each other. They comprise of two components, a discriminator (D), which is trained to differentiate between real generated data; and the generator (G), which creates fake data to fool the discriminator. The generator G takes in a random noise vector Z and parameter vector θ_g and learns to generate synthetic samples very similar to the real data. The resulting $G(Z; \theta_g)$ network is used to generate fake data with data distribution F_g .

Training GAN's on real-world datasets is notoriously hard. One has to carefully tune both generator and discriminator network parameters at training time to avoid conditions like mode collapse and vanishing gradients. Both these conditions affect the quality of synthetic data generated. Choice of loss functions and GAN architectures also play a vital role in avoiding these problems [65].

Tabular Generators. In this work, we deal in the feature space, which is tabular in nature. The values of features are continuous, ordinal, categorical, textual, and/or multi-categorical values. Textual values can be of variable length, high-dimensional due to long-tail distributions of observational data. Textual data is converted into a set of sequences and projected into a trained embedding that preserves semantic meaning similar to methods proposed in Natural Language Processing (NLP) literature. Traditional Generative Adversarial Network (GAN) [40] architectures proposed in literature perform poorly on tabular data [66] due to – (a) Real data often consists of mixed data types. To generate this data simultaneously, a GAN must be able to apply both softmax and tanh on the output; (b) Images pixel values generally follow a Gaussian-like distribution, which can be normalized to $[-1, 1]$, whereas with tabular data, continuous data often have non-Gaussian distributions with long tails that leads to gradient saturation; continuous data type and imbalanced categorical columns need preprocessing steps before feeding into generators.

Recently proposed Conditional Tabular GAN or CTGAN [66] addresses some of the key problems for tabular data synthesis. The training process involves simulating records one by one. It first randomly selects one of the variables, then it randomly selects a value for that variable. Given that value for that variable, the algorithm finds a matching row from the training data. It also generates the rest of the variables conditioning on the selected variable. The generated and true rows are sent to the critic that gives a score. Similarly, modified WGAN-GP [68] changed the generator of the WGAN-GP. Figure 1 summarizes the CTGAN and

WGAN-GP training procedures. to handle continuous variables. Other recently proposed Robust Variational AutoEncoders (R-VAEs) [67], which inherits the properties of typical Variational AutoEncoders, handles the tabular data more efficiently.

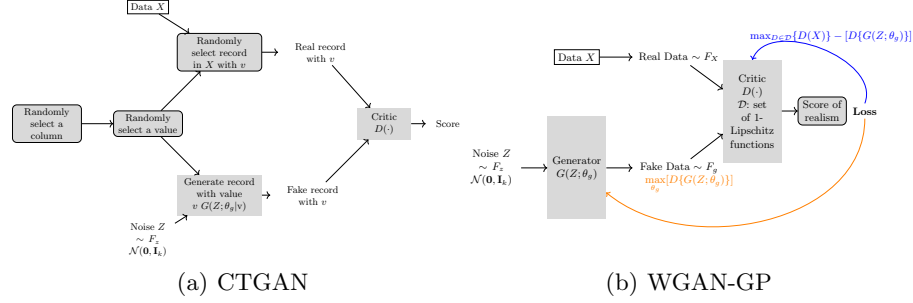


Fig. 1: CTGAN and WGAN-GP Training dynamics

DPGAN [52] and PATEGAN [61] are popular differentially private GAN's. DPGAN adds noise to the gradient of the discriminator during training to create differential privacy guarantees. PATEGAN first trains multiple teacher models on disjoint distributions of data. At inference time, to classify a new sample it noisily aggregates outputs from previously trained teacher models to generate the final output.

3 Instance level Privacy Scoring

To avoid MIA type attacks and unintentional leakage of sensitive attributes, we devise an instance level privacy score for generated synthetic data. As shown in previous studies, the main reasons for privacy compromises in generative models is mainly because of over fitting of training data that could lead to data memorization. We empirically measure *memorization coefficient* α_m for each generated sample and use this measure as privacy score. When distributing synthetic data to external third parties, users can filter samples with high privacy score and reduce privacy risks.

Given X_r , X_s real and synthetic data set generated by some generative algorithm Φ respectively. We first convert into latent vectors in embedding space $E(X_r)$, $E(X_s)$. Next, we measure the spherelet distance between each real and generated sample latent vectors. The spherelet distance between $s(X_i)$ and $s(X_j)$ is defined as:

$$d_S = d_R(X_i, X_j) + d_E(X_i, X_j) \quad (3)$$

where d_E is the Euclidean distance between two sets of points, and d_R is Riemannian divergence (geometric) of dataset of X_i and X_j . The intuition behind this is

we are measuring the distance between points by projecting the samples on to a sphere centered at c with radius r . Spherelet of X denoted by $s(X) = S(V, c, r)$ where V determines an affine subspace the sphere lies in. The spherical error [60] of X is defined by

$$\epsilon(X) = \frac{1}{n} \sum_{i=1}^n (\|x_i - c\| - r)^2 \quad (4)$$

If x_i lie on the sphere then $\epsilon(X) = 0$. Also the distance metric captures (a) which points are found in data manifold in low probability; (b) how close the synthetic and real in nearest neighbour proximity; which can give us a proxy measure for over-fitting and data memorization. *memorization coefficient* α_m for a generated sample X_{g_i} is defined by g_{n_i}/t_{n_i} where t_{n_i} and g_{n_i} are sample counts from training and generated dataset, which have proximity distance lesser than a threshold $> \lambda$. In short, we are measuring the *support set* for the sample X_{g_i} in the training and synthetic set. Samples, which have $\alpha_m < 1$ have large influence of training set and are prone to MI attacks vs samples with $\alpha_m > 1$ capture the underlying distribution of training data without leaking private training set attributes.

The advantage of this method is the model Φ is agnostic and works on data level. Users can discard the samples which cross a certain threshold privacy score to protect privacy of users and, data audit by compliance bodies can be performed without a need to know the model internals and training dynamics.

4 MIA Method

Membership Inference The goal of a membership inference attack is to create a function that accurately predicts whether a data point belongs to the training set of the classifier. To achieve this, an attacker trains a set of surrogate models [73] on confidence scores produced by the target model and membership as labels to achieve her goal.

Typically, the attacker exploits the model confidence reflecting overfitting to infer membership. [73] first demonstrated MIA with only black-box access to the confidence scores for each input data sample. Their approach involved first training many 'surrogate' ML models similar to the target model and uses these shadow models to train a membership inference model to identify whether a sample was in the training set of any of the shadow models. MIA is evaluated as a binary classification task, where metrics like false-negative, false-positive rates, and F1 scores reflect the attack success rates of predicting that a given sample is part of the training set.

We adapt 'shadow' training model in our MI attack on synthetic data. Given an auxiliary dataset $\mathcal{D}_{aux}$ similar to the training data distribution, we first randomly sample single data point $\mathbf{L}_t$ and remove it from the dataset. Now, a well-known ensemble learning method Bootstrap aggregating (Bagging) is used to divide remaining dataset $\mathcal{D}_{aux} - 1$ into subsamples with replacement. Generative models are trained on each sub sample and using the trained models syn_0

synthetic datasets of size k is sampled. These samples are labeled $l_j = 0$ indicating that synthetic data does not contain target sample $\mathbf{L}_t$. Same procedure is repeated by adding $\mathbf{L}_t$ into the dataset $\mathcal{D}_{aux}$ and the synthetic data sets syn_1 of size k is sampled. These samples are labeled $l_j = 1$ indicating that synthetic data contain target sample $\mathbf{L}_t$. Now we have labeled dataset that captures the membership relationship and we train a random forest classifier MI_c , which predicts the probability of a given sample is member or not of the original data set. Given a synthetic dataset S_{Aux} , MI_c predicts the membership of the target record t in the training set of the generative model G that produced S_{Aux} .

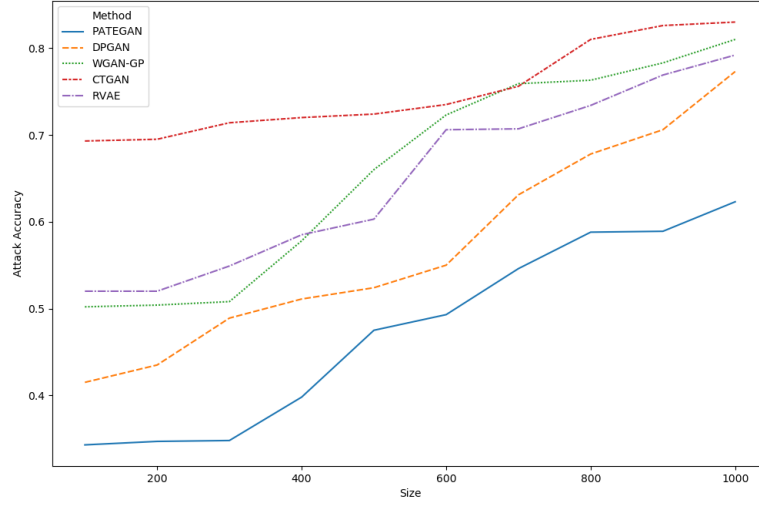
5 Experiments

The membership privacy leakage is evaluated through a membership privacy experiment [48, 49]. Given a training set D_t which follows a data distribution $\mathcal{D}$ and a learning algorithm ϕ is trained on D_t . We assume the adversary has only access to ϕ via a query interface, which exposes inference and class probability of a given sample. In synthetic data sharing use cases, the adversary has access to synthetic datasets generated by some generative model. Here the goal of the adversary is to discover the membership of a given sample i.e. it belongs to training data or not with access to only synthetic data. We assume domain-dependent data access to the adversary. In our experiments, to measure the strength of the proposed Instance Level Privacy filtering of synthetic data we measure the accuracy drop of MI attack with/without filtering mechanism.

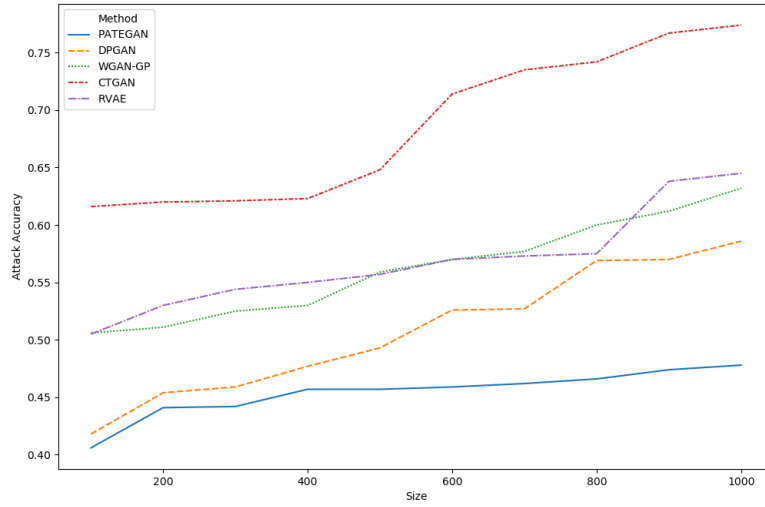
For this work, we use two datasets – First dataset is SIVEP-Gripe database [2] released by Brazilian government with records of 99,557 ICU Covid-19 patients in Brazil, which includes personal information (ethnicity, age and location). A logistic regression (binary classification) model is fit to predict patient-level COVID-19 mortality with 6882 data sample and 25 features. Second dataset is the Brazilian E-Commerce Public Dataset by Olist⁴. It consists of 112,000 customer orders at multiple marketplaces in Brazil. The dataset contains order status, price, payment and freight performance to customer location, product attributes and finally reviews written by customers. Customer and Seller Id’s are anonymized, and references to the companies and partners in the review text have been replaced with the names of Game of Thrones great houses. For our experiment we use columns price, customer identifier, customer zip code, customer city as feature columns. A logistic regression model is fit to predict the customer spend by aggregating price column.

We create 5 synthetic datasets using CTGAN, WGAN-GP, PATEGAN, DP-GAN, and, RVAE as synthetic data generators for SIVEP-Gripe and E-Commerce data. The training of generator the input dimensions of architecture is adjusted to accommodate the dataset shape. All hidden layers use the ReLU activation function, the final layer uses the sigmoid function. The discriminator network also uses three hidden layers, which are symmetric to the generator. The hidden layers again use ReLU. We train each network with 100

⁴ <https://www.kaggle.com/olistbr/brazilian-ecommerce/>

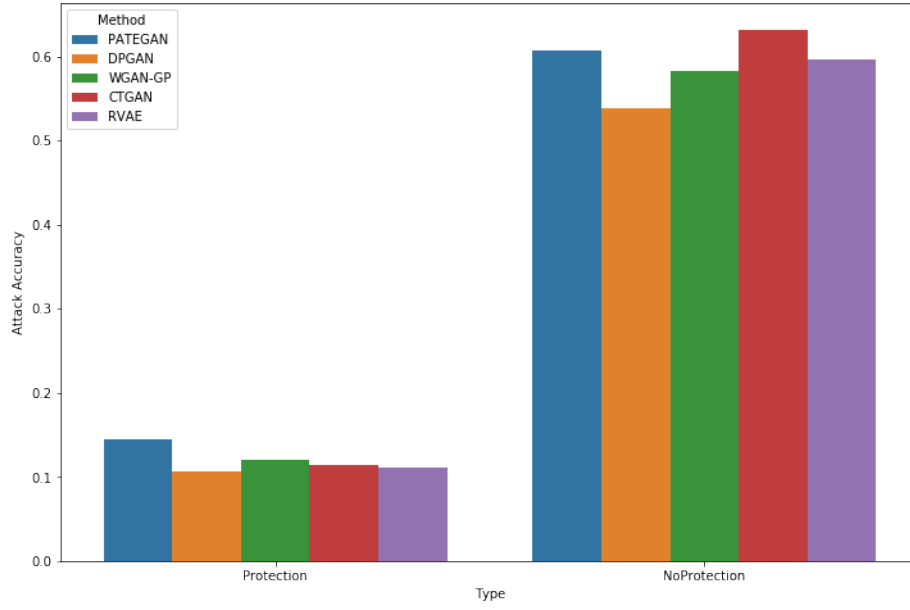


(a)

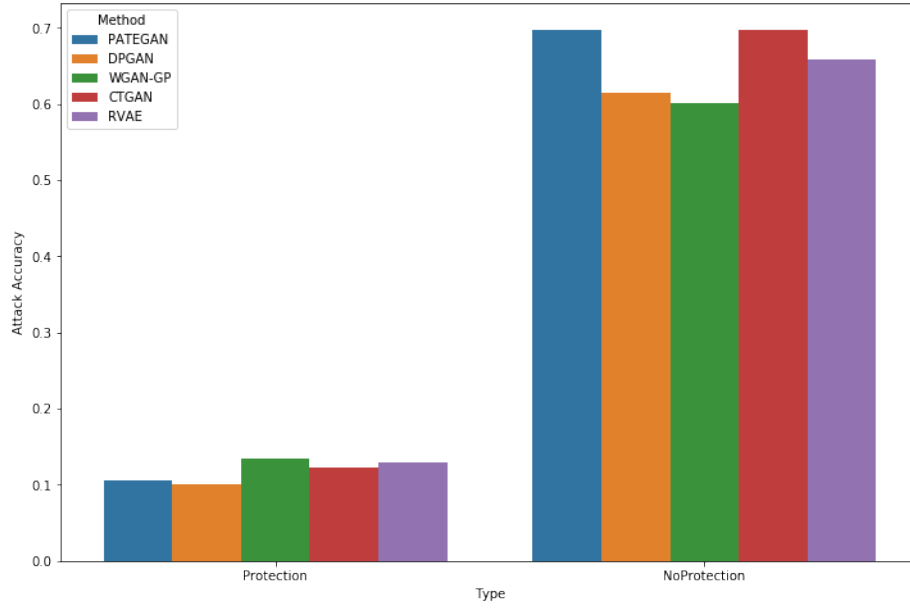


(b)

Fig.2: Synthetic data size dependency on MIA attack for (a) E-Commerce Dataset (b) EHR dataset



(a)



(b)

Fig. 3: Attack Accuracy drop before and after filtering data points with high privacy scores - (a) EHR dataset (b) E-Commerce Dataset

epochs. For training differential private GAN’s, we used the privacy budgets of $\epsilon = [0.01, 0.1, 0.5, 1.0, 3.0]$. λ is set to 0.5 to reflect the adversary advantage in membership attack. Randomly 1000 samples are sampled from training and marked as $\mathcal{D}_{aux}$. To reflect maximum compromise by adversary as white box setting, we use the same generator to train the shadow model $\mathcal{S}$.

5.1 Experimental Results and Discussion

We first present experimental results of the membership inference attack and the attack accuracy drop before and after we filter the data with α_m method. We measure the accuracy of attack with respect to number of samples in the auxiliary dataset used in the attack mechanism. Figure 5.1 and 3 summarizes the privacy risk of each synthesising method in terms of MI attack accuracy and auxiliary dataset. We observe that increasing the size of the synthetic dataset increases the risk of privacy leakage. As more information about the data is learnt by adversary the privacy of the data generator weakens. Similarly, differential privacy methods that add noise to the output with some privacy budget ϵ , the attack accuracy decreases showing the value of DP-based generators reduce privacy risks. Increase in privacy strength of the dataset is consistent across datasets and different models highlighting the strength of our proposed method.

5.2 Discussion

To develop good synthetic data generation models we still have some technical hurdles, for example, it is not only hard to exactly predict what data characteristics will be preserved in a model’s stochastic output but also appropriate metrics to empirically compare two distributions are lacking. The privacy budgets in terms of perturbations to achieve differential privacy make it even harder to predict which records will remain vulnerable, and might even increase the exposure of some data records. Our work aims to give some direction towards improving synthetic data privacy by assigning instance-level privacy scores. A data auditor or service provider can leverage the scores to remove the data points prone to MI attacks. The advantage of our method is a model agnostic and it can be used in post-hoc fashion. We explored the accuracy drop from the attacker end, but measuring the accuracy-privacy tradeoff due to dropping the data points from the synthetic data on downstream models needs further investigation.

We want to highlight some of the limitations of our work. We only tested our method on tabular data, further investigation is needed to understand the privacy risks of other types of data such as text, images and, audio. Additionally, it can be observed that we use the whole dataset to measure α_m , which may be misleading as pointed out by [46], membership leakage for subgroup may be more vulnerable than others for discriminative models. We plan to address this drawback in our future work. It is also important to point out that the privacy risk measures provided in this analysis are dataset-dependent and *can* be generator-specific. While the proposed method gives some empirical privacy

guarantees for generative models and guidance for data sharing use case, we need more detailed analysis on the dependency of α_m against model parameters, and varied dataset distributions. In terms of threat models, our analysis does not take into consideration an attacker who has knowledge of the filtering scheme in advance. A malicious attacker can use only filtered data points to train surrogate models making the proposed scheme invalid.

6 Conclusion

One of the major impediments in advancing ML in security-sensitive domains is the lack of open realistic publicly available datasets. To gain trust and protect the end-user data and, comply with regulations stakeholders are leveraging synthetic data as a tool to share data between third-party aggregators and researchers. However, the privacy analysis of these datasets is still open research. In this work, we fill this gap by investigating the privacy issues of the generated data sets from the attacker and auditor point of view. We propose an Instance-level Privacy Score (PS) for each synthetic sample by measuring the memorization coefficient α_m per sample. Leveraging, PS we empirically show that the accuracy of membership inference attacks on synthetic data drops significantly. PS is a model agnostic, post-training measure, which helps data sharer with guidance about the privacy properties of a given sample but also helps third party data auditors to run privacy checks without sharing model internals. Finally, there is a lot more work to be done purely within the realm of privacy and our work only addresses a very small issue empirically. Understanding social and legal implications of synthetic data sharing needs inputs and partnerships from policy experts and, compliance officers.

References

1. Carlini, Nicholas and Tramer, Florian and Wallace, Eric and Jagielski, Matthew and Herbert-Voss, Ariel and Lee, Katherine and Roberts, Adam and Brown, Tom and Song, Dawn and Erlingsson, Ulfar and others
Extracting Training Data from Large Language Models , arXiv preprint arXiv:2012.07805
2. SIVEP-Gripe. <http://plataforma.saude.gov.br/coronavirus/dados-abertos/>. In *Ministry of Health. SIVEP-Gripe public dataset*, (accessed May 10, 2020; in Portuguese), 2020.
3. N. P. Jouppi, C. Young, N. Patil, D. Patterson, G. Agrawal, R. Bajwa, S. Bates, S. Bhatia, N. Boden, A. Borchers, et al. In-datacenter performance analysis of a tensor processing unit. In *Proceedings of the 44th Annual International Symposium on Computer Architecture*, pages 1–12, 2017.
4. Departement of Commerce, National Institute of Standards and Technology, “Differential private synthetic data challenge.” <https://www.challenge.gov/challenge/differential-privacy-synthetic-data-challenge/>, 2019. Accessed 2021-02-19.
5. T. Theraux, Olivier, “Anonymisation and synthetic data: towards trustworthy data.” <https://theodi.org/article/>

- anonymisation-and-synthetic-data-towards-trustworthy-data/, 2019. Accessed 2021-02-19.
6. The Open Data Institute, “Diagnosing the nhs: Synae.” <https://www.odileeds.org/events/synae/>. Accessed 2021-02-19.
 7. Rubin, D. B. Discussion: Statistical Disclosure Limitation, 1993.
 8. Rubin, D. B. Multiple imputation after 18+ years. *American Statistical Association*, 91(434):473–489, 1996.
 9. “Hazy.” <https://hazy.com/>.
 10. “Aireverie.” <https://aireverie.com/>.
 11. “Statice.” <https://statice.ai/>.
 12. “one-view.” <https://one-view.ai/>.
 13. “datagen.” <https://www.datagen.tech/>.
 14. “Synthesize.” <https://synthesize.io/>.
 15. “cognata.” <https://www.cognata.com/>.
 16. “Mostly-ai.” <https://mostly.ai/>.
 17. D. B. Rubin, “Statistical disclosure limitation,” *Journal of official Statistics*, vol. 9, no. 2, pp. 461–468, 1993.
 18. Little, R. J. Statistical Analysis of Masked Data. *Journal of Official Statistics*, 9 (2):407–426, 1993.
 19. Abowd, J. M. and Lane, J. New Approaches to Confidentiality Protection: Synthetic Data, Remote Access and Research Data Centers. In Domingo-Ferrer, J. and Torra, V. (eds.), *Privacy in Statistical Databases*, volume 3050, pp. 282–289. 2004. ISBN 978-3-540-22118-0.
 20. Abowd, J. M. and Woodcock, S. D. Multiply-Imputing Confidential Characteristics and File Links in Longitudinal Linked Data. In *Privacy in Statistical Databases*, volume 3050, pp. 290–297. 2004. ISBN 3-540-22118-2.
 21. Reiter, J. P. and Raghunathan, T. E. The multiple adaptations of multiple imputation. *Journal of the American Statistical Association*, 102(480):1462–1471, 2007.
 22. Drechsler, J. and Reiter, J. P. Sampling with synthesis: A new approach for releasing public use census microdata. *Journal of the American Statistical Association*, 105(492):1347–1357, 2010.
 23. Drechsler, J. and Reiter, J. P. An Empirical Evaluation of Easily Implemented, Nonparametric Methods for Generating Synthetic Datasets. *Computational Statistics and Data Analysis*, 55(12):3232–3243, dec 2011.
 24. Kinney, S. K., Reiter, J. P., Reznick, A. P., Miranda, J., Jarmin, R. S., and Abowd, J. M. Towards Unrestricted Public Use Business Microdata: The Synthetic Longitudinal Business Database. *International Statistical Review*, 79(3):362–384, 2011.
 25. Reiter, J. P. Using cart to generate partially synthetic, public use microdata. *Journal of Official Statistics*, 21(3):441–462, 2005a.
 26. Caiola, G. and Reiter, J. P. Random Forests for Generating Partially Synthetic, Categorical Data. *Transactions on Data Privacy*, 3(1):27–42, 2010.
 27. Drechsler, J. *Synthetic Datasets for Statistical Disclosure Control*, volume 53. Springer, 2011. ISBN 9788578110796.
 28. Bowen, C. M. and Liu, F. Comparative Study of Differentially Private Data Synthesis Methods. *Statistical Science*, forthcoming.
 29. Drechsler, J. *Synthetic Datasets for Statistical Disclosure Control*, volume 53. Springer, 2011. ISBN 9788578110796.
 30. Manrique-Vallier, D. and Hu, J. Bayesian Non-Parametric Generation of Fully Synthetic Multivariate Categorical Data in the Presence of Structural Zeros. *Journal of the Royal Statistical Society. Series A: Statistics in Society*, 181(3):635–647, 2018.

31. Snoke, J., Raab, G., Nowok, B., Dibben, C., and Slavkovic, A. General and specific utility measures for synthetic data. 2016.
32. Raghunathan, T. E., Reiter, J. P., and Rubin, D. B. Multiple imputation for statistical disclosure limitation. *Journal of official statistics*, 19(1):1, 2003.
33. Kinney, S. K., Reiter, J. P., Reznick, A. P., Miranda, J., Jarmin, R. S., and Abowd, J. M. Towards Unrestricted Public Use Business Microdata: The Synthetic Longitudinal Business Database. *International Statistical Review*, 79(3):362–384, 2011.
34. Kinney, S. K., Reiter, J. P., and Berger, J. O. Model Selection When Multiple Imputation is Used to Protect Confidentiality in Public Use Data. *Journal of Privacy and Confidentiality*, 2(2):3–19, 2010.
35. Article 29 Data Protection Working Party - European Commission, “Opinion 05/2014 on anonymisation techniques.” https://ec.europa.eu/justice/article-29/documentation/opinion-recommendation/files/2014/wp216_en.pdf, 2014.
36. M. Elliot, E. Mackey, K. O’Hara, and C. Tudor, *The anonymisation decision-making framework*. UKAN Manchester, 2016.
37. I. S. Rubinstein and W. Hartzog, “Anonymization and risk,” *Wash. L. Rev.*, vol. 91, p. 703, 2016.
38. M. Elliot, K. O’Hara, C. Raab, C. M. O’Keefe, E. Mackey, C. Dibben, H. Gowans, K. Purdam, and K. McCullagh, “Functional anonymisation: Personal data and the data environment,” *Computer Law & Security Review*, vol. 34, no. 2, pp. 204–221, 2018.
39. N. Patki, R. Wedge, and K. Veeramachaneni, “The synthetic data vault,” in *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410, IEEE, 2016.
40. I. Goodfellow, “Nips 2016 tutorial: Generative adversarial networks,” *arXiv preprint arXiv:1701.00160*, 2016.
41. M. Elliot, K. O’Hara, C. Raab, C. M. O’Keefe, E. Mackey, C. Dibben, H. Gowans, K. Purdam, and K. McCullagh, “Functional anonymisation: Personal data and the data environment,” *Computer Law & Security Review*, vol. 34, no. 2, pp. 204–221, 2018.
42. European Commission (2016). *Regulation (EU) 2016/679: General Data Protection Regulation (GDPR)*.
43. L. Sweeney, “k-anonymity: A model for protecting privacy,” *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 05, pp. 557–570, 2002.
44. R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership Inference Attacks Against Machine Learning Models,” in *IEEE Symposium on Security and Privacy (S&P)*, 2017.
45. A. Machanavajjhala, J. Gehrke, D. Kifer, and M. Venkatasubramanian, “L-diversity: privacy beyond k-anonymity,” in *22nd International Conference on Data Engineering (ICDE’06)*, 2006.
46. M. Yaghini, B. Kulynych, and C. Troncoso, “Disparate vulnerability: On the unfairness of privacy attacks against machine learning,” *arXiv preprint arXiv:1906.00389*, 2019.
47. H. Ping, J. Stoyanovich, and B. Howe, “Datasyntesizer: Privacy-preserving synthetic datasets,” in *Proceedings of the 29th International Conference on Scientific and Statistical Database Management*, 2017.
48. B. Jayaraman, L. Wang, D. Evans, and Q. Gu, “Revisiting membership inference under realistic assumptions,” *arXiv preprint arXiv:2005.10881*, 2020.

49. S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pp. 268–282, IEEE, 2018.
50. J. Zhang, G. Cormode, C. M. Procopiuc, D. Srivastava, and X. Xiao, "Privbayes: Private data release via bayesian networks," *ACM Trans. Database Syst.*, 2017.
51. L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional gan," in *Advances in Neural Information Processing Systems*, 2019.
52. Yoon, J., Drumright, L. N., and Van Der Schaar, M. Anonymization through data synthesis using generative adversarial networks (ads-gan). *IEEE Journal of Biomedical and Health Informatics*, 2020.
53. Brock, A., Donahue, J., and Simonyan, K. Large scale gan training for high fidelity natural image synthesis. *arXiv preprint arXiv:1809.11096*, 2018.
54. Adlam, B., Weill, C., and Kapoor, A. Investigating under and overfitting in wasserstein generative adversarial networks. *arXiv preprint arXiv:1910.14137*, 2019.
55. Meehan, C., Chaudhuri, K., and Dasgupta, S. A non-parametric test to detect data-copying in generative models. *arXiv preprint arXiv:2004.05675*, 2020.
56. Jamie Hayes, Luca Melis, George Danezis, and Emiliano De Cristofaro. Logan: Membership inference attacks against generative models. *Proceedings on Privacy Enhancing Technologies*, 2019(1):133–152, 2019.
57. Sablayrolles, A.; Douze, M.; Ollivier, Y.; Schmid, C.; and Jégou, H. 2019. White-box vs Black-box: Bayes Optimal Strategies for Membership Inference. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 5558–5567.
58. A. Sablayrolles, M. Douze, Y. Ollivier, C. Schmid, and H. Jégou, "White-box vs black-box: Bayes optimal strategies for membership inference," 2019.
59. S. Truex, L. Liu, M. E. Gursoy, L. Yu, and W. Wei, "Towards demystifying membership inference attacks," *ArXiv*, vol. abs/1807.09173, 2018.
60. Kuppa, Aditya and Grzonkowski, Slawomir and Asghar, Muhammad Rizwan and Le-Khac, Nhien-An Black Box Attacks on Deep Anomaly Detectors, Proceedings of the 14th International Conference on Availability, Reliability and Security
61. J. Yoon, J. Jordon, and M. van der Schaar. PATE-GAN: Generating synthetic data with differential privacy guarantees. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=S1zk9iRqF7>.
62. Torkzadehmahani, R., Kairouz, P., and Paten, B. DP-CGAN: differentially private synthetic data and label generation. *CoRR*, abs/2001.09700, 2020.
63. D. Arpit, S. Jastrzundzki, N. Ballas, D. Krueger, E. Bengio, M. S. Kanwal, T. Maharaj, A. Fischer, A. Courville, Y. Bengio, and et al., "A closer look at memorization in deep networks," in *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML'17, p. 233–242, JMLR.org, 2017.
64. S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pp. 268–282, July 2018.
65. Salimans, T., Goodfellow, I.J., Zaremba, W., Cheung, V., Radford, A., Chen, X.: Improved techniques for training gans. *CoRR* **abs/1606.03498** (2016)
66. Xu, L., Skoularidou, M., Cuesta-Infante, A., Veeramachaneni, K.: Modeling tabular data using conditional gan. In: *NeurIPS* (2019)
67. Eduardo, S., Nazabal, A., Williams, C.K.I., Sutton, C.: Robust Variational Autoencoders for Outlier Detection and Repair of Mixed-Type Data. In: *Proceedings of AISTATS* (2020)
68. Camino, R., Hammerschmidt, C., State, R., 2018. Generating multi-categorical samples with generative adversarial networks. *arXiv preprint arXiv:1807.01202*.

69. C. R. Meehan, K. Chaudhuri, and S. Dasgupta, “A non-parametric test to detect data-copying in generative models,” *ArXiv*, vol. abs/2004.05675, 2020.
70. Z. Izzo, M. A. Smart, K. Chaudhuri, and J. Zou, “Approximate data deletion from machine learning models: Algorithms and evaluations,” *ArXiv*, vol. abs/2002.10077, 2020.
71. Congzheng Song and Vitaly Shmatikov. Overlearning Reveals Sensitive Attributes. In *International Conference on Learning Representations (ICLR)*, 2020.
72. Luca Melis, Congzheng Song, Emiliano De Cristofaro, and Vitaly Shmatikov. Exploiting Unintended Feature Leakage in Collaborative Learning. In *IEEE Symposium on Security and Privacy (S&P)*, pages 497–512. IEEE, 2019.
73. Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership Inference Attacks Against Machine Learning Models. In *IEEE Symposium on Security and Privacy (S&P)*, pages 3–18. IEEE, 2017.
74. Min Chen, Zhikun Zhang, Tianhao Wang, Michael Backes, Mathias Humbert, and Yang Zhang. When Machine Unlearning Jeopardizes Privacy. *CoRR abs/2005.02205*, 2020.
75. Zheng Li and Yang Zhang. Label-Leaks: Membership Inference Attack with Label. *CoRR abs/2007.15528*, 2020.
76. Klas Leino and Matt Fredrikson. Stolen Memories: Leveraging Model Memorization for Calibrated White-Box Membership Inference. In *USENIX Security Symposium (USENIX Security)*, pages 1605–1622. USENIX, 2020.
77. Dingfan Chen, Ning Yu, Yang Zhang, and Mario Fritz. GAN-Leaks: A Taxonomy of Membership Inference Attacks against Generative Models. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 343–362. ACM, 2020.
78. Ahmed Salem, Yang Zhang, Mathias Humbert, Pascal Berrang, Mario Fritz, and Michael Backes. ML-Leaks: Model and Data Independent Membership Inference Attacks and Defenses on Machine Learning Models. In *Network and Distributed System Security Symposium (NDSS)*. Internet Society, 2019.
79. Jinyuan Jia, Ahmed Salem, Michael Backes, Yang Zhang, and Neil Zhenqiang Gong. MemGuard: Defending against Black-Box Membership Inference Attacks via Adversarial Examples. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 259–274. ACM, 2019.